

Project Title: EAL Tech SOC Home Lab – WAZUH01 Build Log

Date: May 23 ,2026

Author: Ezekiel Libara

Server Details:

VM Name: WAZUH01

Operating System: Ubuntu Server 26.04 LTS

RAM: 8GB

Storage: 100GB

Role: SIEM Server - Security Monitoring

Hosted On: VMware Workstation 16 Pro

Software:

SIEM Tool: Wazuh 4.7.5

Dashboard Access: <https://192.168.73.129>

Default User: admin

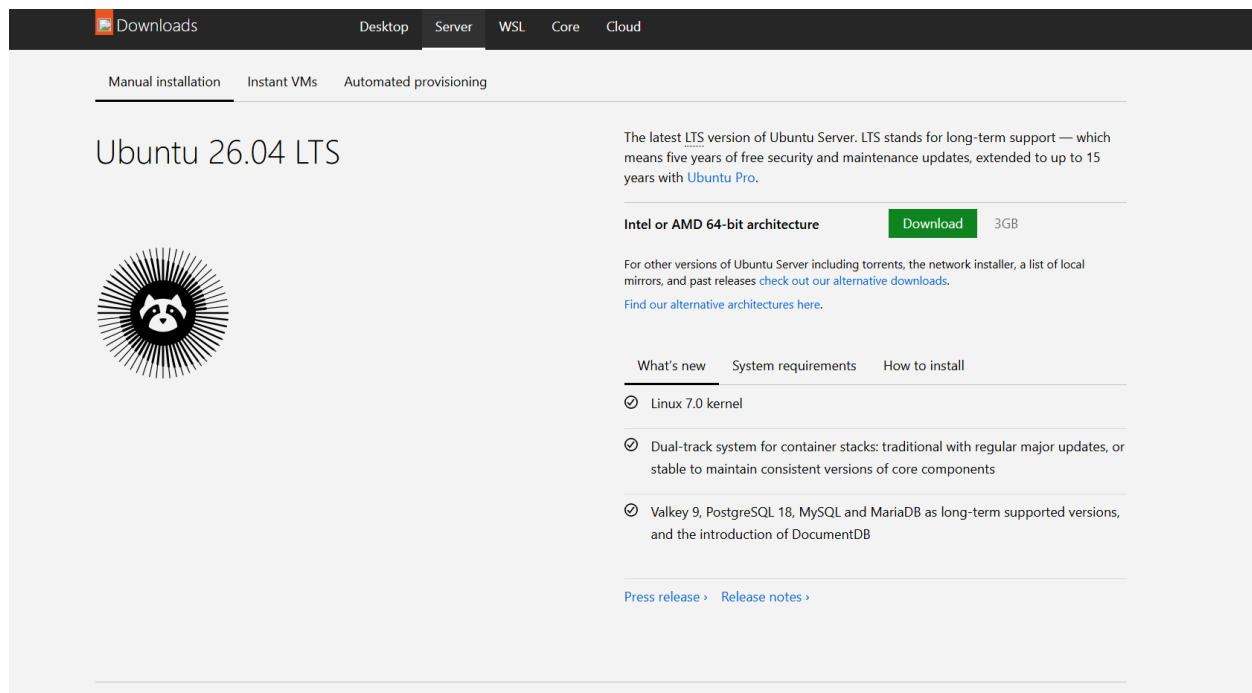
Purpose: This document records the step by step process of building WAZUH01 ,the Security Monitoring server for the EAL Tech SOC Home Lab. WAZUH01 runs Ubuntu Server and Wazuh SIEM to monitor and detect security threats across the home lab network.

Build Steps:

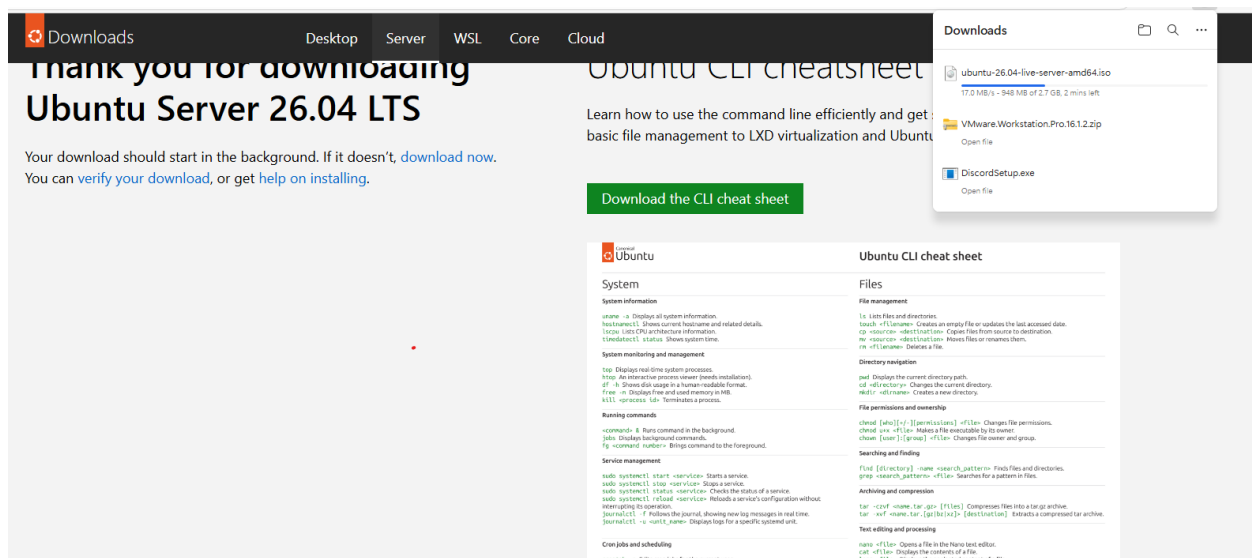
Step 1 — Download Ubuntu Server ISO

- Downloaded Ubuntu Server 26.04 LTS

- Source: <https://ubuntu.com/download/server>



Step 2 : Then click the green "Download" button



File size: 3GB

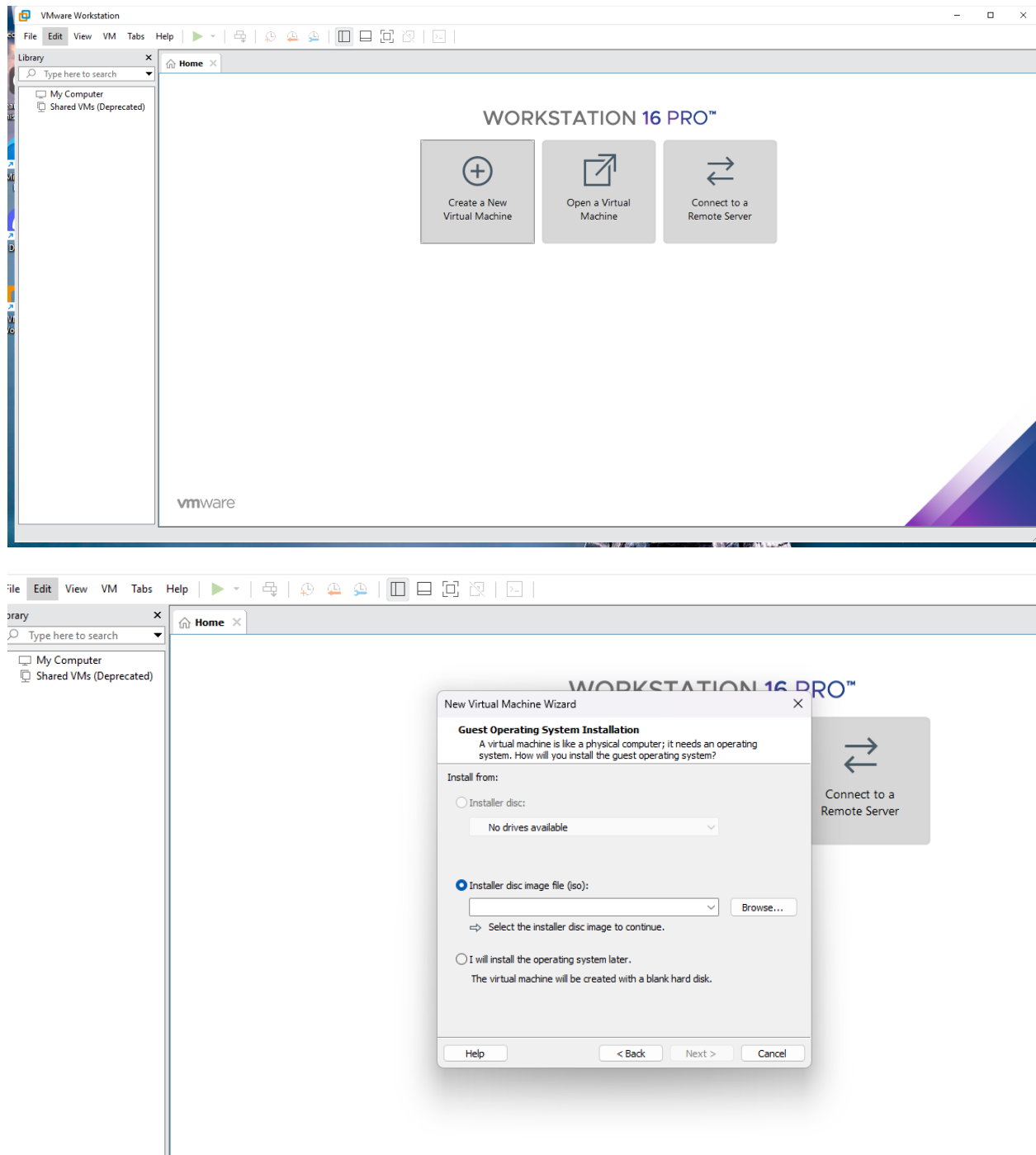
Version: Ubuntu Server 26.04 LTS

Download date: May 23, 2026

Step 3 — Open VMware and Create New VM

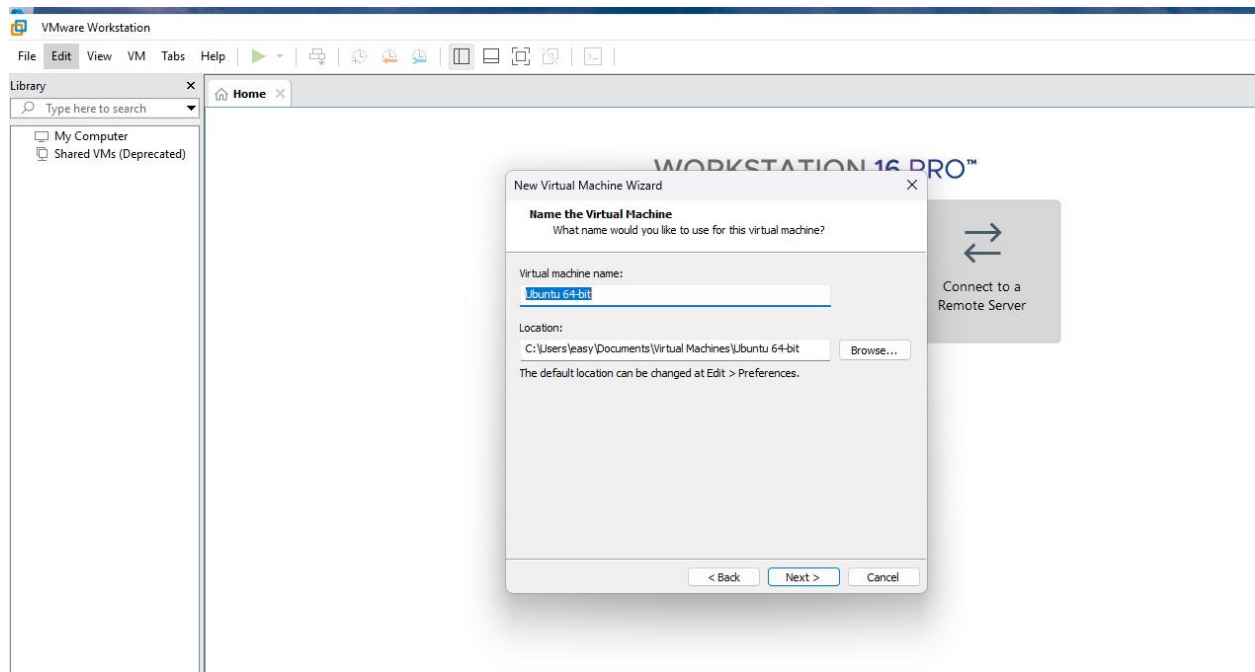
Opened VMware Workstation 16 Pro

Clicked Create a New Virtual Machine



Step 4 — Selected Ubuntu Server ISO

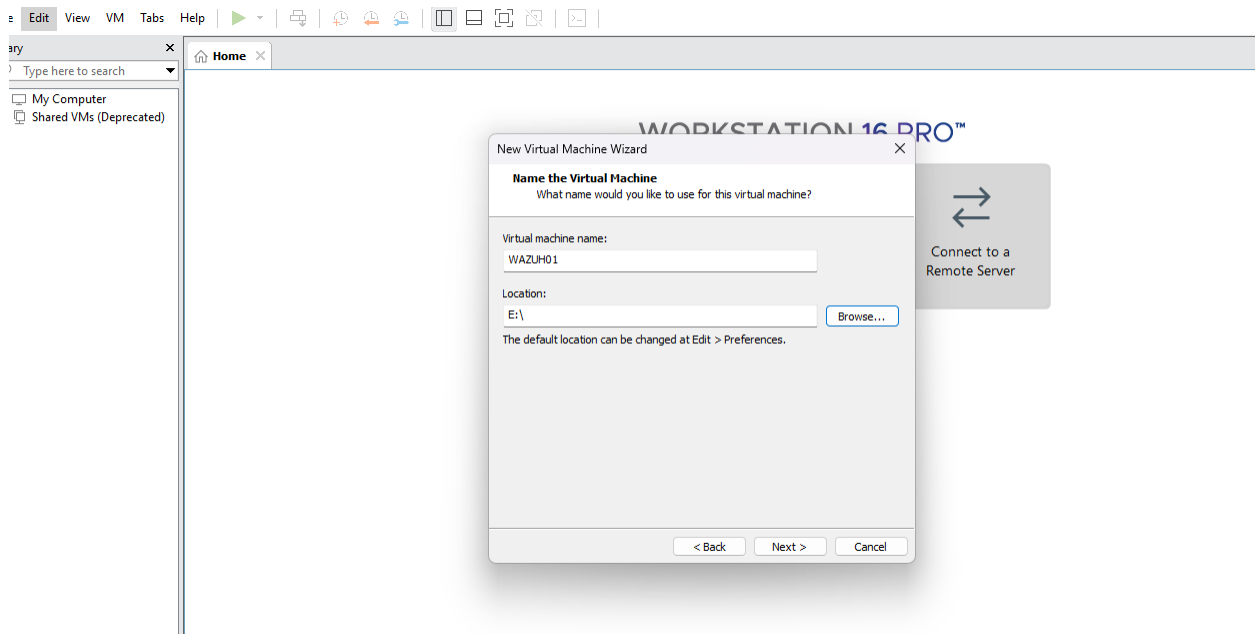
- Selected "Installer disc image file (iso)"
- Clicked Browse to find the ISO file



Step 5 — Named the Virtual Machine

VM Name: WAZUH01

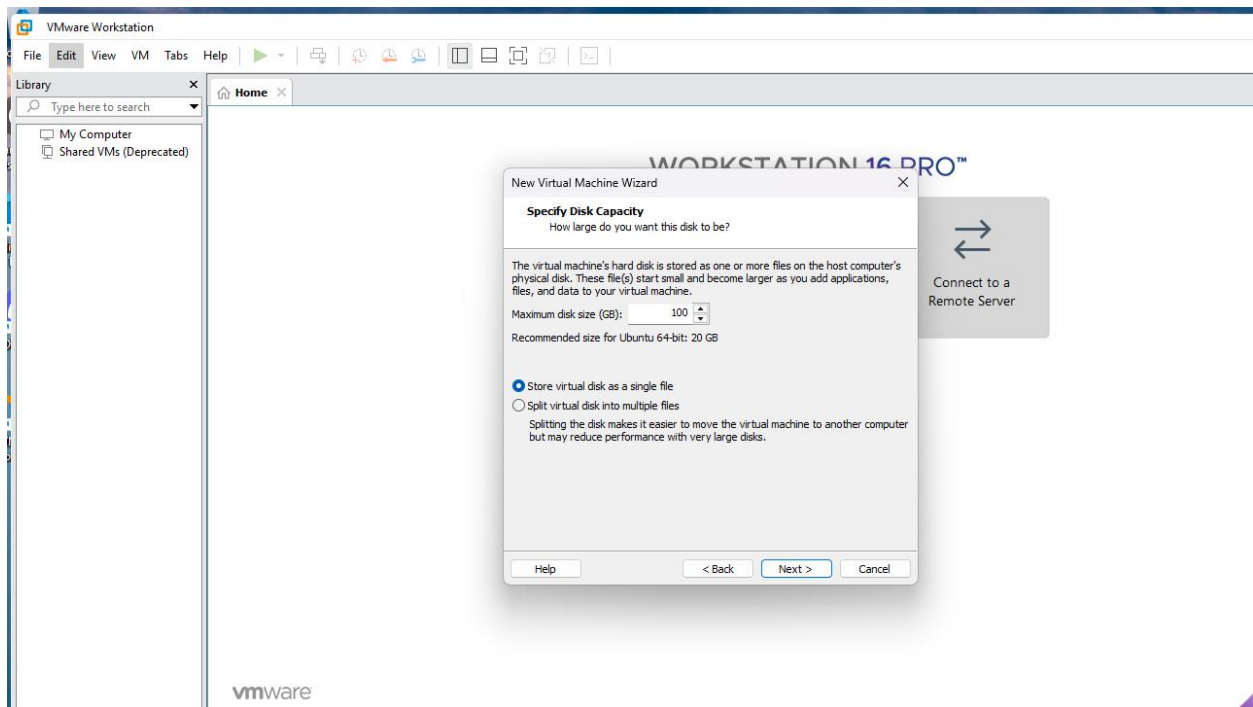
Storage Location: E:\ (Internal VM drive)



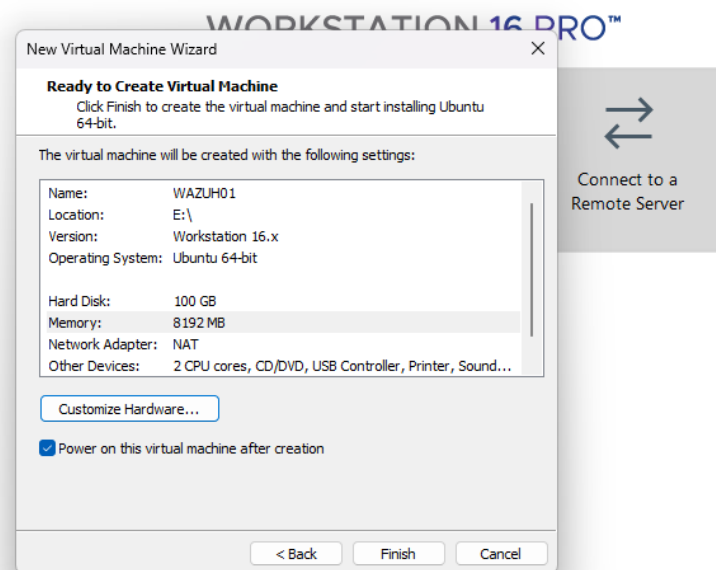
Step 6 — Assigned Disk Storage

Disk size: 100GB

Type: Single file



Click Next :



Step 7 — VM Creation Summary

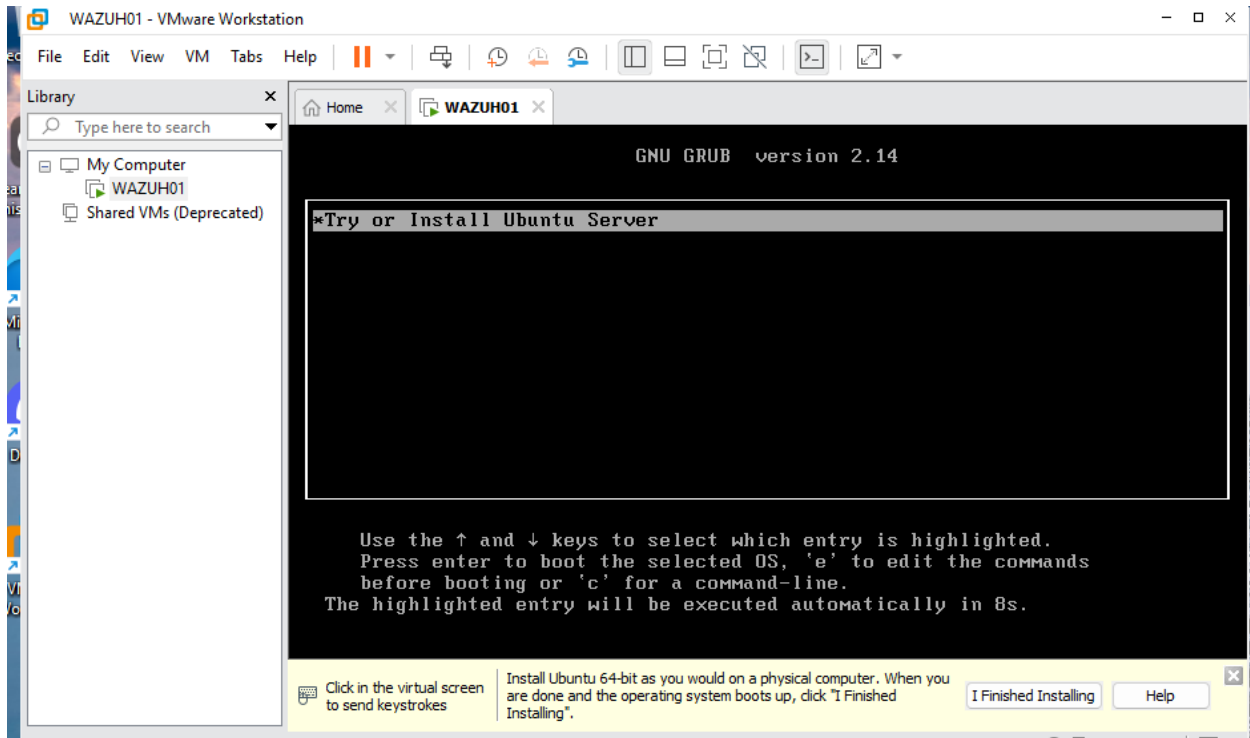
VM Name: WAZUH01

Location: E:\

Operating System: Ubuntu 64-bit

Hard Disk: 100GB

Memory: 8GB (8192MB)
Network: NAT
Result: VM created successfully
Then click Finish

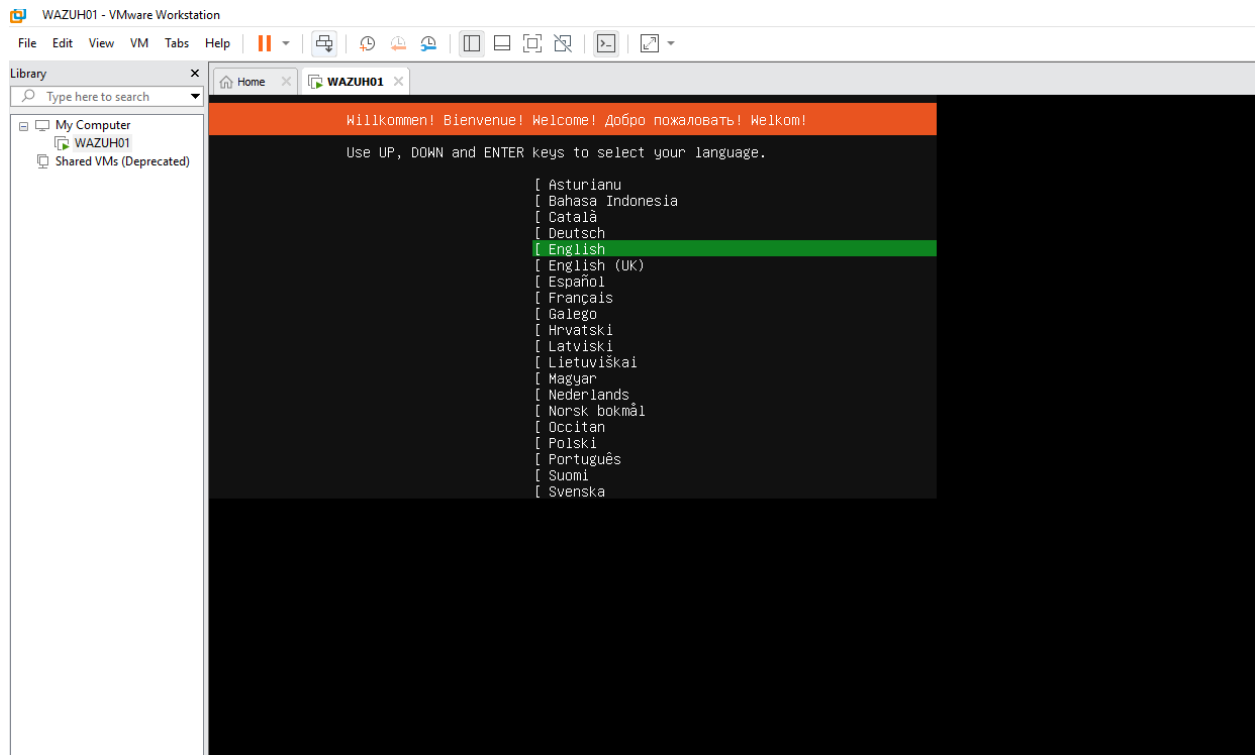


Step 8 — Ubuntu Server Boot Menu

VM WAZUH01 started successfully

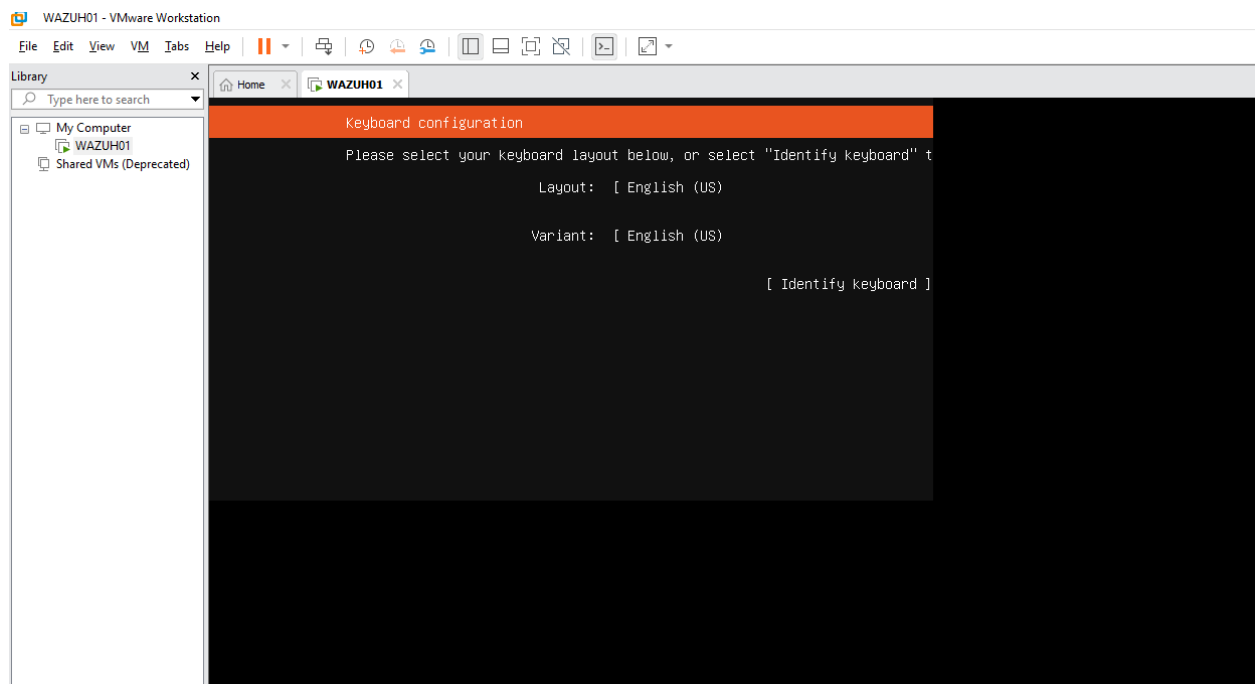
Ubuntu Server boot menu appeared

Selected "Try or Install Ubuntu Server"



Step 9 — Selected Language

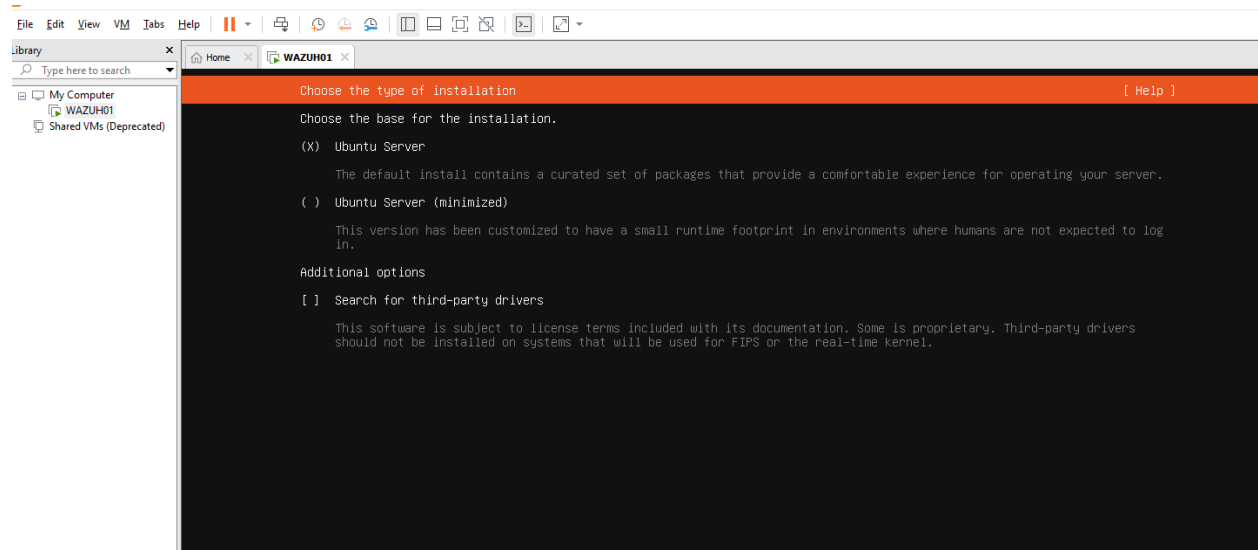
- Selected: English



Step 10 — Keyboard Configuration

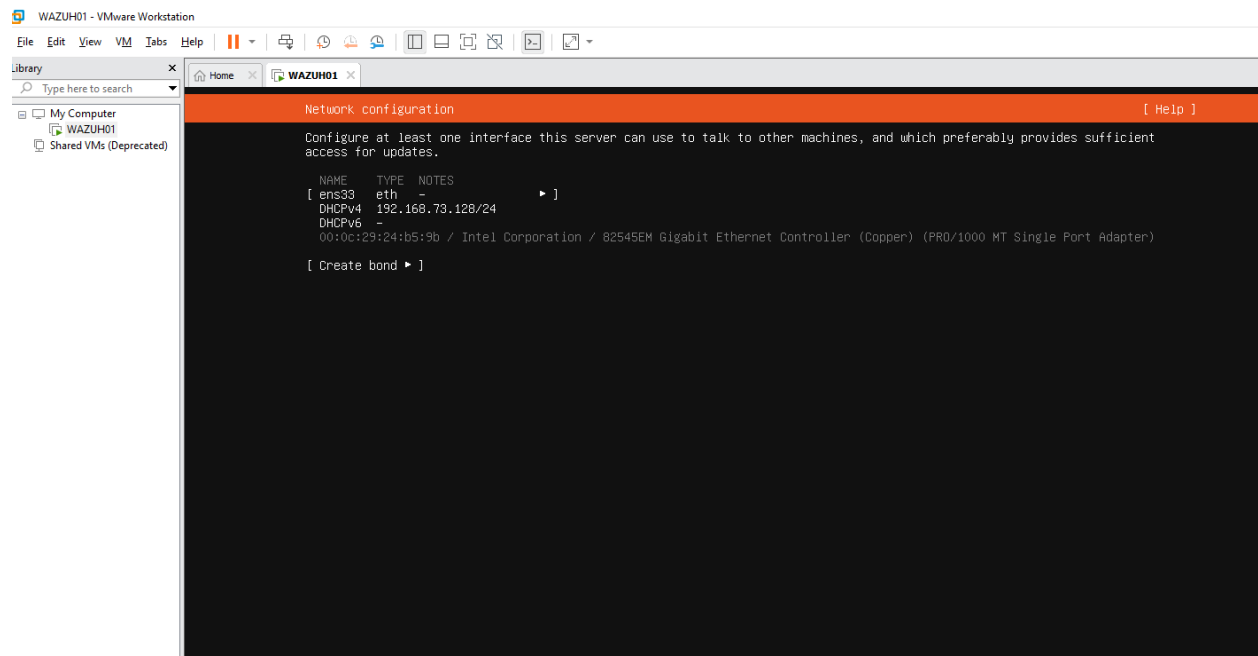
- Layout: English (US)

- Variant: English (US)



Step 11 — Installation Type

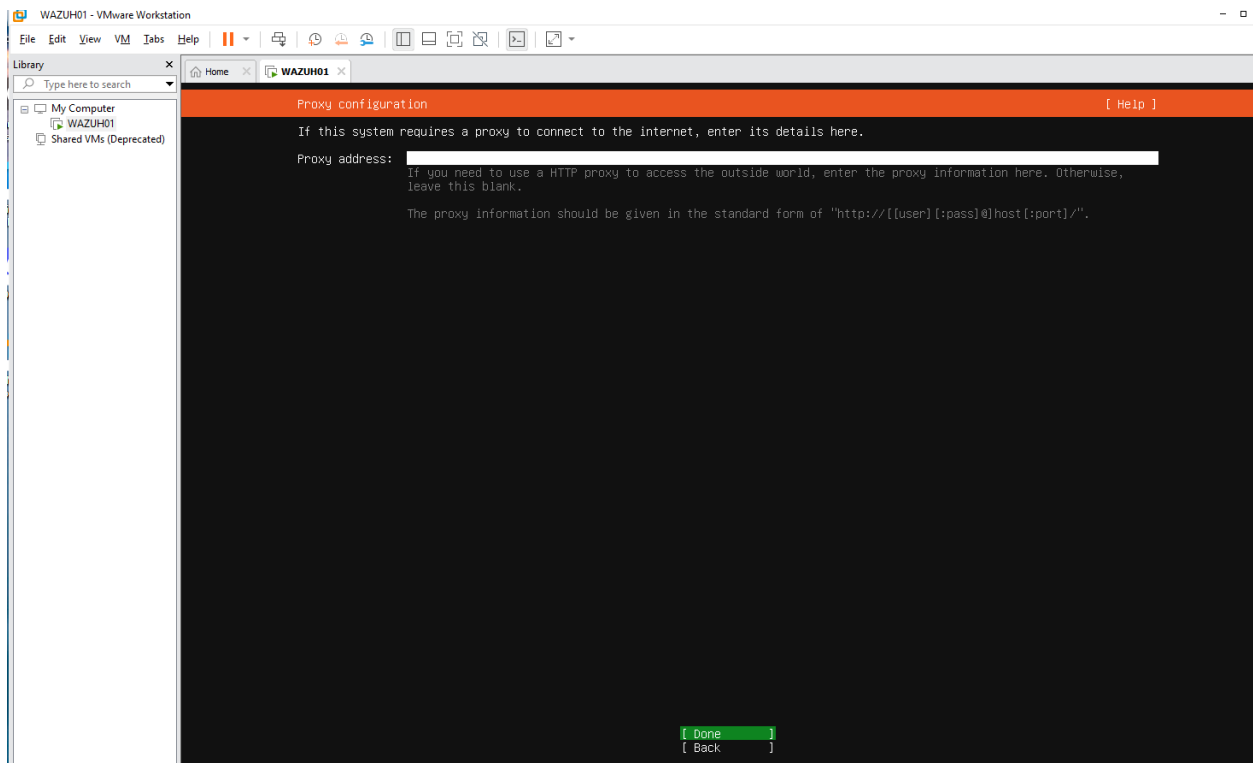
- Selected: Ubuntu Server (default)
- Reason: Full installation needed for Wazuh SIEM



Step 12 — Network Configuration

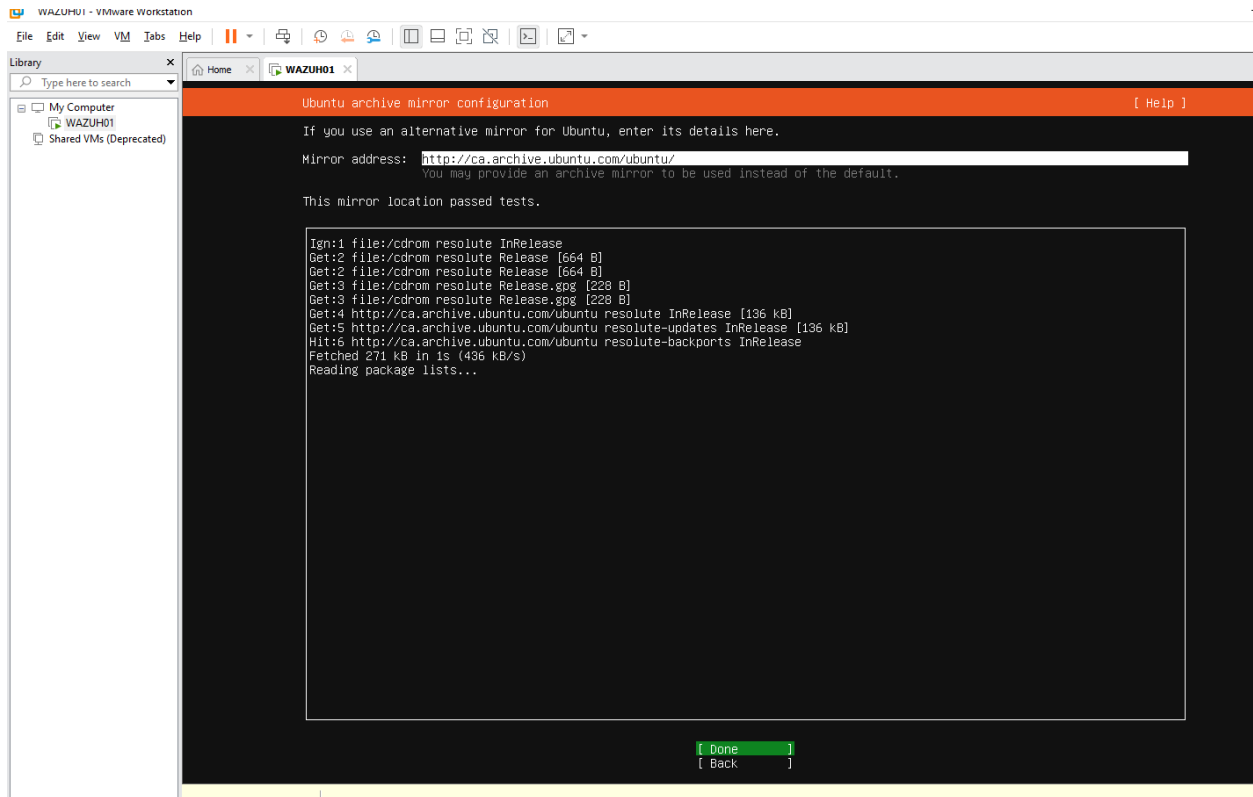
- Interface: ens33

- Type: Ethernet (DHCP)



Step 13 — Proxy Configuration

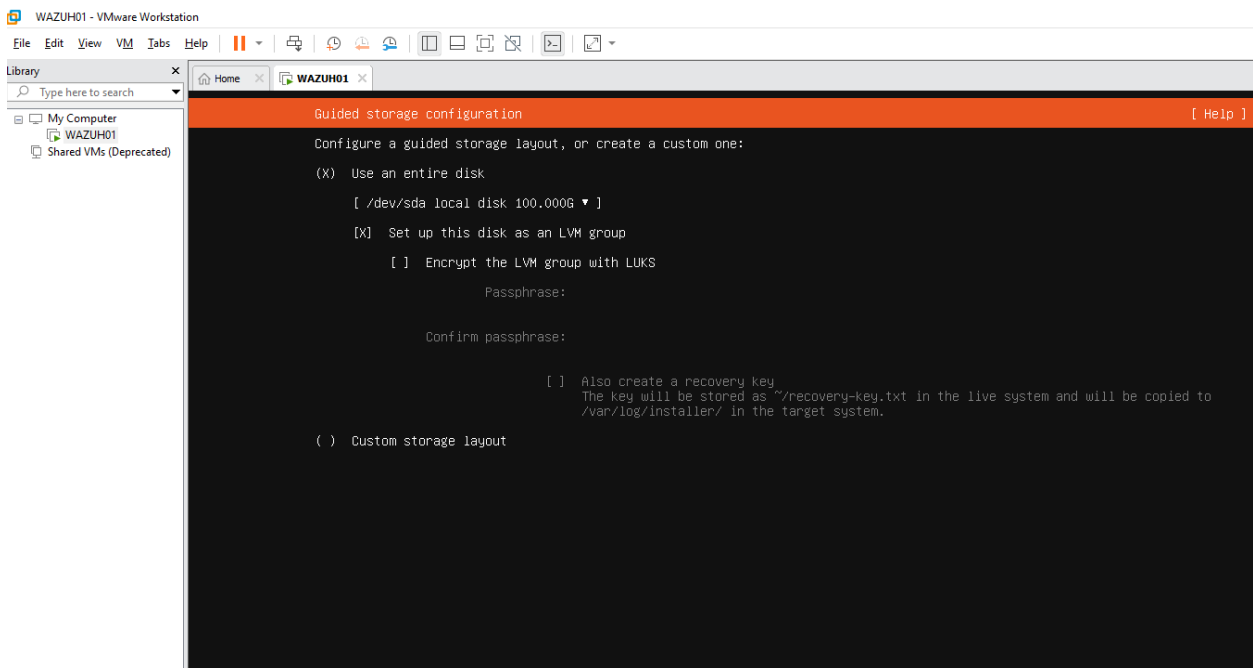
- Proxy address: None (left blank)
- Reason: Home lab does not use a proxy



Step 14 — Ubuntu Mirror Configuration

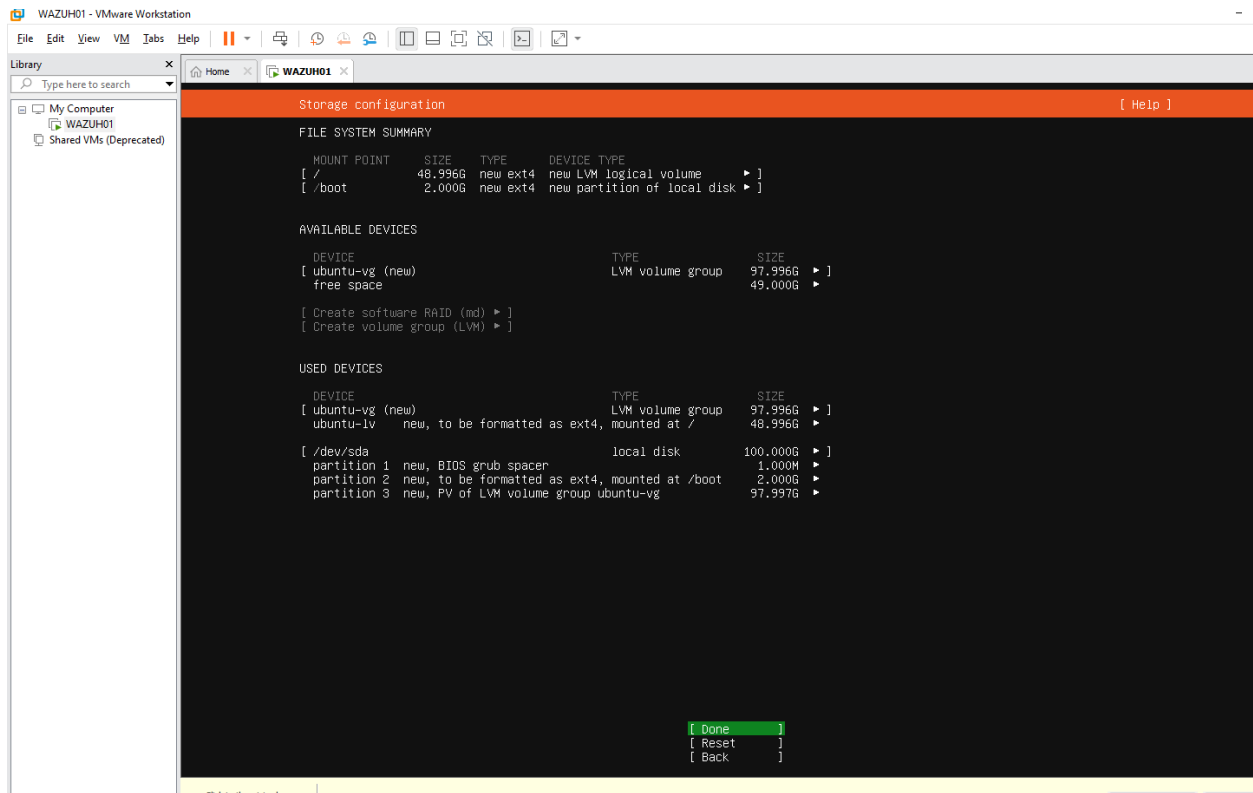
- Mirror: <http://ca.archive.ubuntu.com/ubuntu/>

- Status: Passed tests 



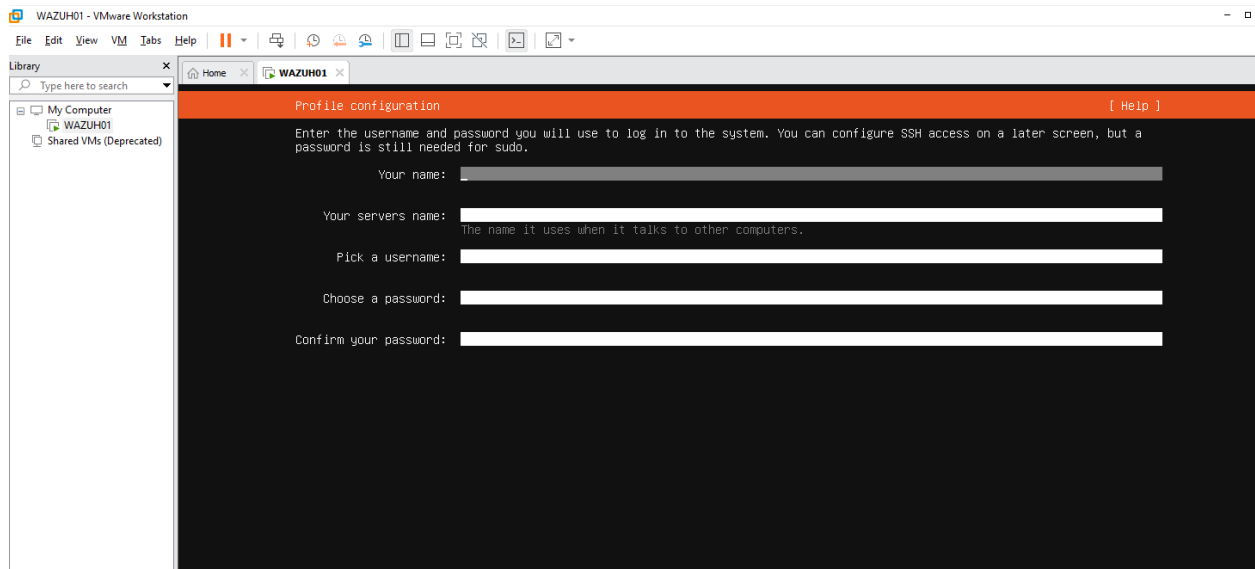
Step 15 — Storage Configuration

- Selected: Use entire disk
- Disk size: 100GB
- LVM: Enabled
- Encryption: Disabled



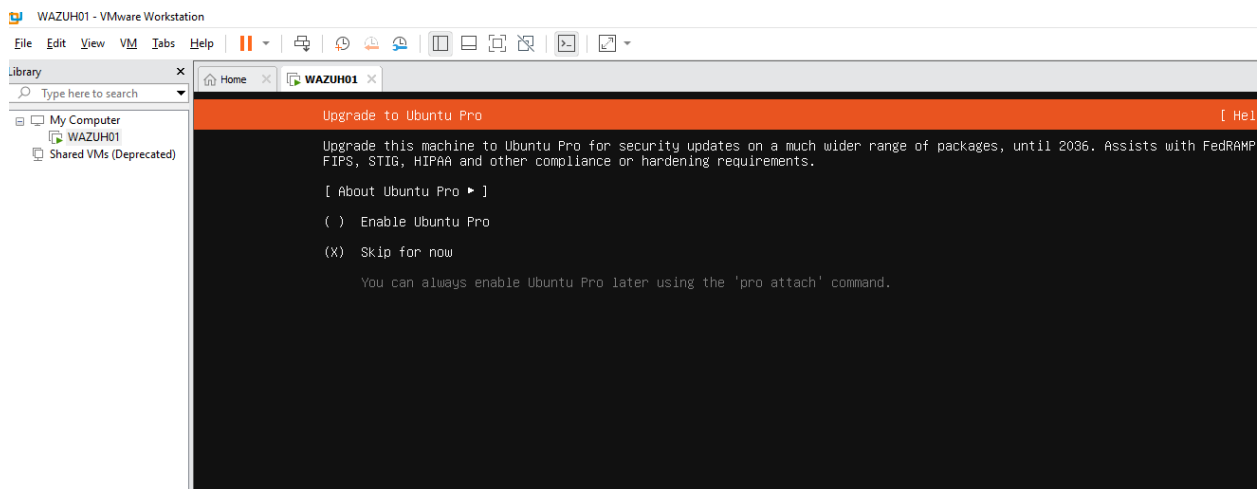
Step 16 — Storage Layout Summary

- Total Disk: 100GB
- Boot partition: 2GB
- LVM Volume: 97GB



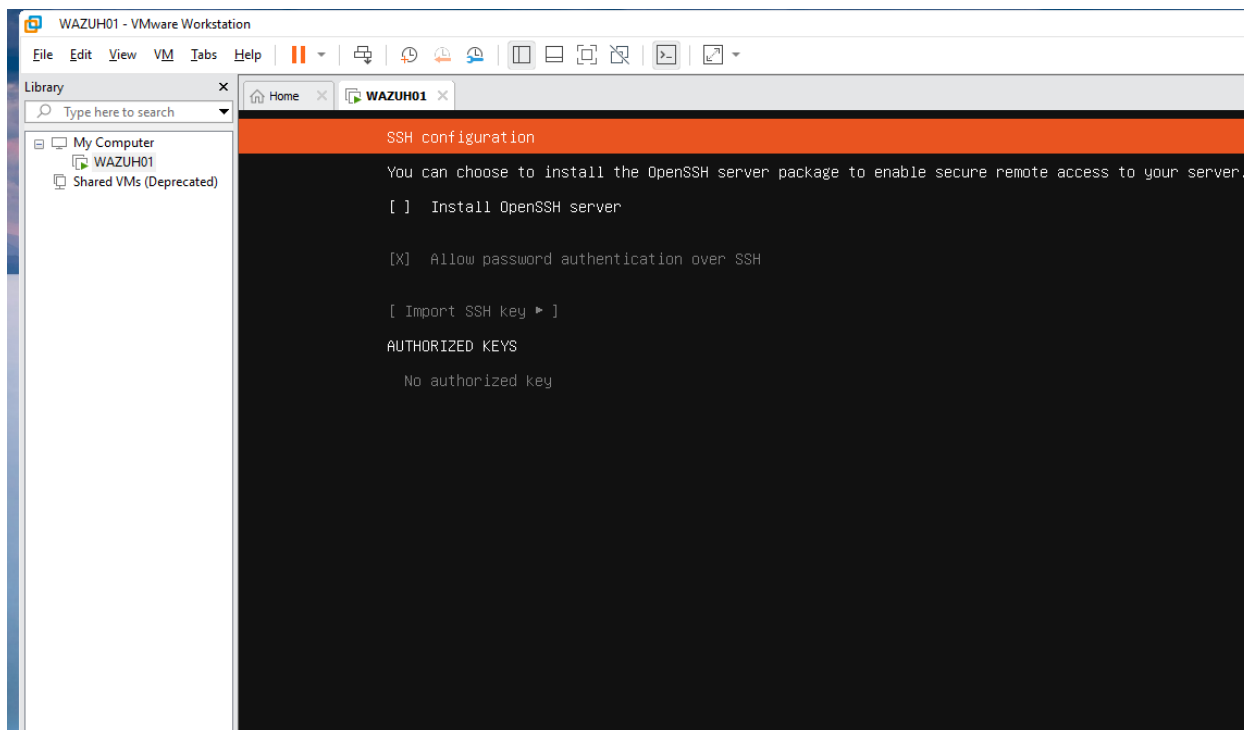
Step 17 — Profile Configuration

- Full name: Ezekiel Libara
- Server name: WAZUH01
- Username: easy



Step 18 — Ubuntu Pro

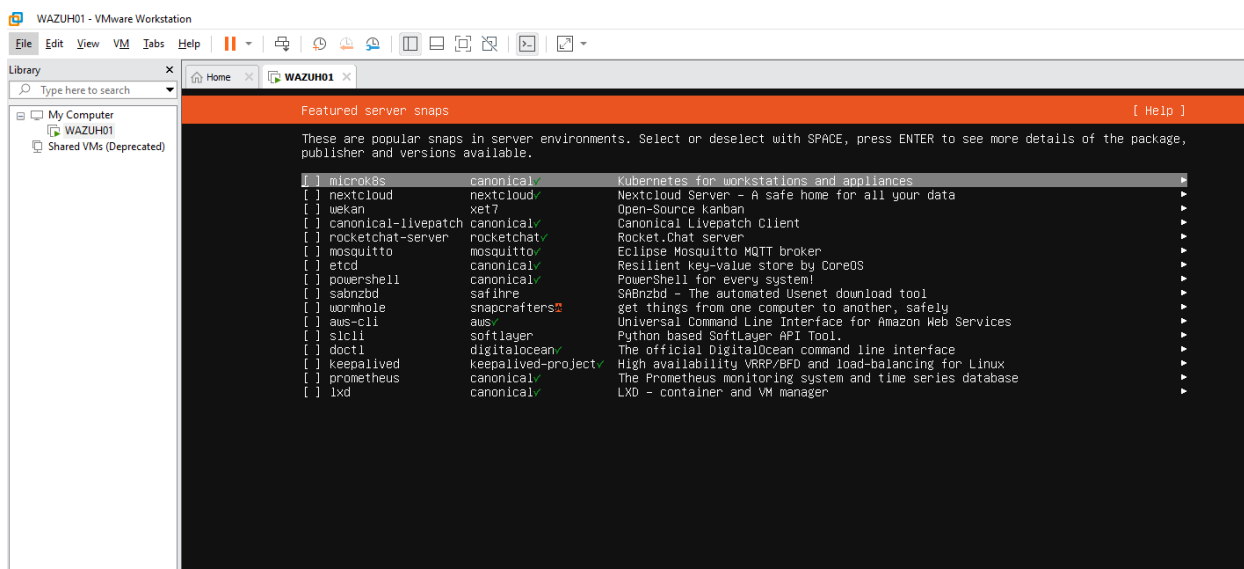
- Selected: Skip for now
- Reason: Not required for home lab



Step 19 — SSH Configuration

Installed: OpenSSH server

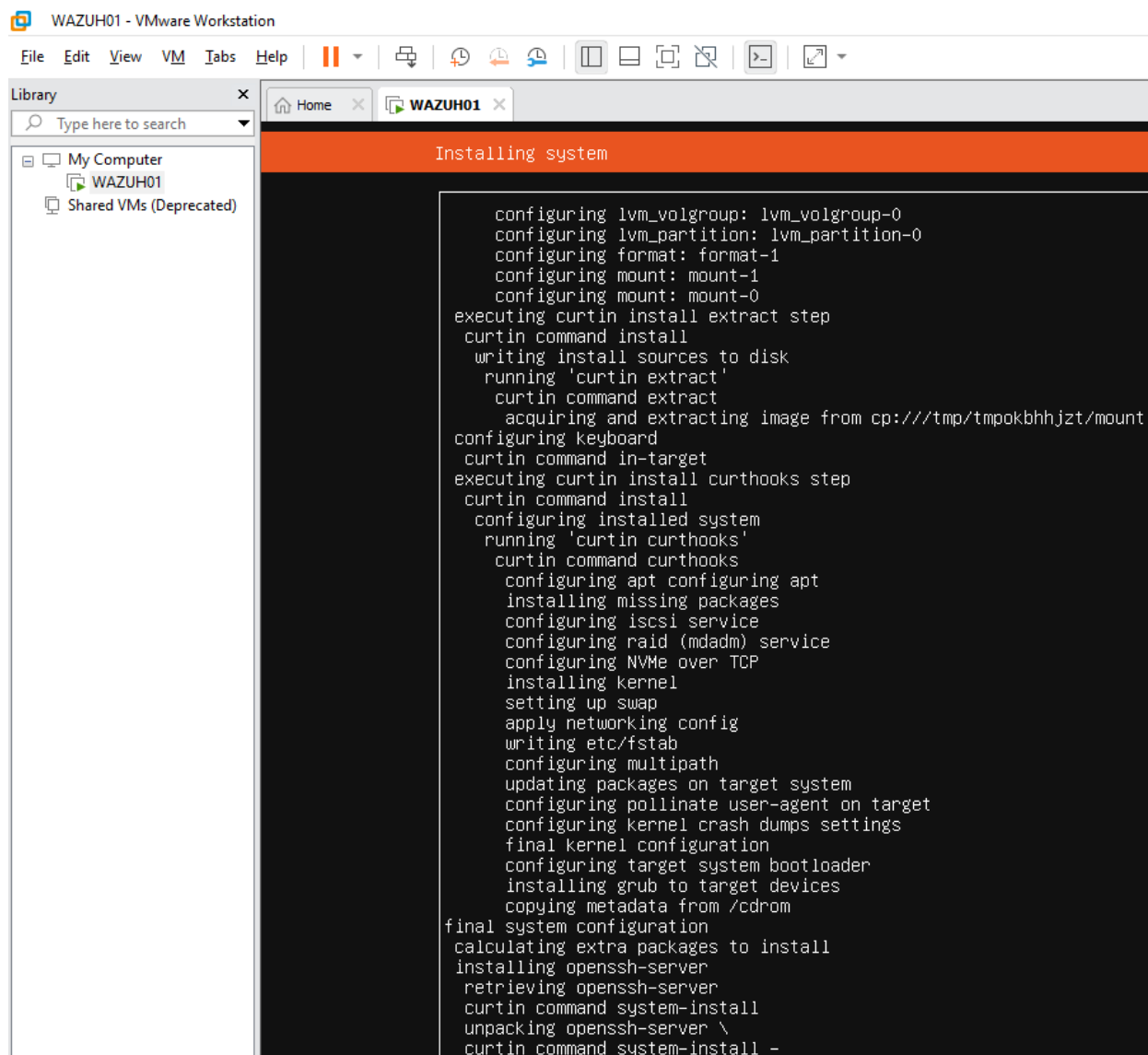
Reason: Allows remote access to WAZUH01



Step 20 — Featured Server Snaps

Selected: None

Reason: Wazuh will be installed manually



Step 21 — Ubuntu Server Installing

System installation started

OpenSSH server being installed

```

[ 17.357219] cloud-init[1440]: | ++ooo. |
[ 17.358459] cloud-init[1440]: +----- [SHA256] -----+

Ubuntu 26.04 LTS wazuh01 tty1

Try contacting this VM's SSH server via 'ssh vsock%505722267' from host.

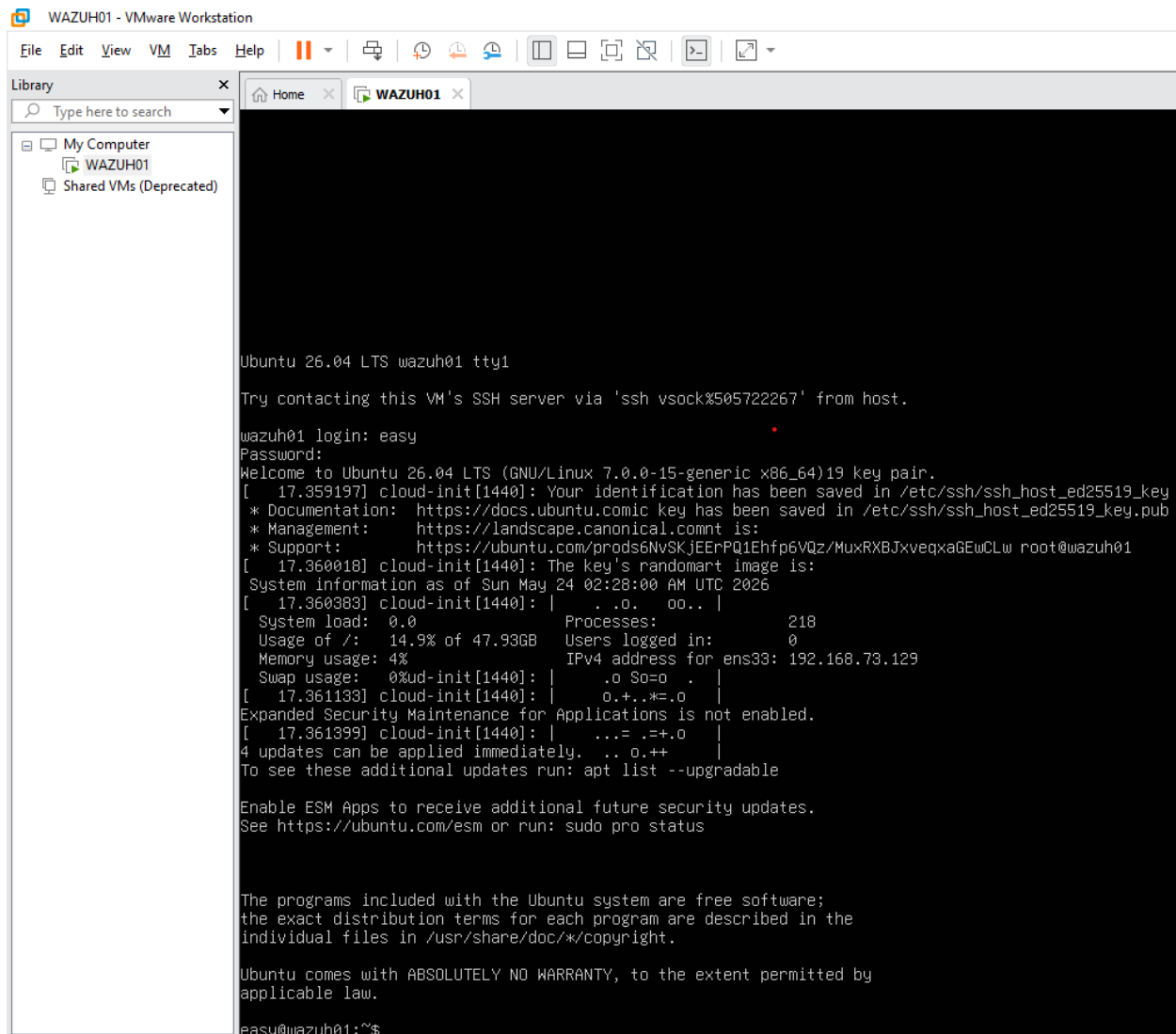
wazuh01 login: _

17.358776] cloud-init[1440]: Generating public/private ed25519 key pair.
[ 17.359197] cloud-init[1440]: Your identification has been saved in /etc
[ 17.359488] cloud-init[1440]: Your public key has been saved in /etc/ssh
[ 17.359689] cloud-init[1440]: The key fingerprint is:
[ 17.359852] cloud-init[1440]: SHA256:bds6NvSKjEErPQ1Ehfp6VQz/MuxRXBJxvec
[ 17.360018] cloud-init[1440]: The key's randomart image is:
[ 17.360168] cloud-init[1440]: +--- [ED25519 256] ---+
[ 17.360383] cloud-init[1440]: | . .o. oo.. |
[ 17.360529] cloud-init[1440]: | +. . ... |
[ 17.360670] cloud-init[1440]: | .+ .+ . o . |
[ 17.360824] cloud-init[1440]: | .E . ++ o . |
[ 17.360995] cloud-init[1440]: | .o So=0 . |
[ 17.361133] cloud-init[1440]: | o.+..*=.o |
[ 17.361267] cloud-init[1440]: | ..=..oo== o |
[ 17.361399] cloud-init[1440]: | ...= .+=.o |
[ 17.361530] cloud-init[1440]: | .. o.++ |
[ 17.361679] cloud-init[1440]: +----- [SHA256] -----+

```

Step 23 — Ubuntu Server Boot Complete

- Server name: wazuh01
- Status: Running and ready for login



```
WAZUH01 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
WAZUH01
Shared VMs (Deprecated)

Ubuntu 26.04 LTS wazuh01 tty1

Try contacting this VM's SSH server via 'ssh vsock%505722267' from host.

wazuh01 login: easy
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-15-generic x86_64)19 key pair.
[ 17.359197] cloud-init[1440]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
* Documentation: https://docs.ubuntu.com key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
* Management: https://landscape.canonical.comnt is:
* Support: https://ubuntu.com/prods6NvSKjEErPQ1Ehfp6VQz/MuxRXBJxveqxaGEwCLw root@wazuh01
[ 17.360018] cloud-init[1440]: The key's randomart image is:
System information as of Sun May 24 02:28:00 AM UTC 2026
[ 17.360383] cloud-init[1440]: | . .o. oo.. |
System load: 0.0 Processes: 218
Usage of /: 14.9% of 47.93GB Users logged in: 0
Memory usage: 4% IPv4 address for ens33: 192.168.73.129
Swap usage: 0%ud-init[1440]: | .o So=o . |
[ 17.361133] cloud-init[1440]: | o.+..*=.o |
Expanded Security Maintenance for Applications is not enabled.
[ 17.361399] cloud-init[1440]: | ...= .+=.o |
4 updates can be applied immediately. .. o.++ |
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

easy@wazuh01:~$
```

Step 24 — Successfully Logged In

- Username: easy
- Server: wazuh01
- IP Address: 192.168.73.129
- Status: Login successful 


```

linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14
se 'sudo apt autoremove' to remove them.

Upgrading:
  base-files distro-info-data motd-news-config snapd

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 0
  Download size: 41.9 MB
  Space needed: 639 kB / 41.1 GB available

Get:1 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 motd-news-config all 14ubuntu6.1 [5,632 B]
Get:2 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu6.1 [75.0 kB]
Get:3 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 distro-info-data all 0.68ubuntu0.1 [5,854 B]
Get:4 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 snapd amd64 2.75.2+ubuntu26.04.2 [41.8 MB]
Fetched 41.9 MB in 3s (13.7 MB/s)
Reading database... 139606 files and directories currently installed.)
Preparing to unpack .../motd-news-config_14ubuntu6.1_all.deb...
Unpacking motd-news-config (14ubuntu6.1) over (14ubuntu6)...
Preparing to unpack .../base-files_14ubuntu6.1_amd64.deb...
Unpacking base-files (14ubuntu6.1) over (14ubuntu6)...
Setting up base-files (14ubuntu6.1)...
motd-news.service is a disabled or a static unit not running, not starting it.
Reading database... 139606 files and directories currently installed.)
Preparing to unpack .../distro-info-data_0.68ubuntu0.1_all.deb...
Unpacking distro-info-data (0.68ubuntu0.1) over (0.68build1)...
Preparing to unpack .../snapd_2.75.2+ubuntu26.04.2_amd64.deb...
Unpacking snapd (2.75.2+ubuntu26.04.2) over (2.74.1+ubuntu26.04.4)...
Setting up motd-news-config (14ubuntu6.1)...
Setting up snapd (2.75.2+ubuntu26.04.2)...
Installing new version of config file /etc/apparmor.d/usr.lib.snapd.snap-confine.real...
snapd.failure.service is a disabled or a static unit not running, not starting it.
snapd.gpio-chardev-setup.target is a disabled or a static unit not running, not starting it.
snapd.snap-repair.service is a disabled or a static unit not running, not starting it.
Setting up distro-info-data (0.68ubuntu0.1)...
Processing triggers for plymouth-theme-ubuntu-text (24.004.60+git20250831.4a3c171d-0ubuntu8)...
Processing triggers for dbus (1.16.2-2ubuntu4)...
Processing triggers for install-info (7.2-5ubuntu2)...
Processing triggers for plymouth (24.004.60+git20250831.4a3c171d-0ubuntu8)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-15-generic

Progress: [ 94%] |

```

Step 26 — Upgrade System Packages

- Command: `sudo apt upgrade -y`
- Purpose: Installs the latest security updates

WAZUH01 - VMware Workstation

File Edit View VM Tabs Help

Library

Type here to search

My Computer

WAZUH01

Shared VMs (Deprecated)

Home WAZUH01

```
Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 0
Download size: 41.9 MB
Space needed: 639 kB / 41.1 GB available

Get:1 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 motd
Get:2 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base
Get:3 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 dist
Get:4 http://ca.archive.ubuntu.com/ubuntu resolute-updates/main amd64 snap
Fetched 41.9 MB in 3s (13.7 MB/s)
(Reading database... 139606 files and directories currently installed.)
Preparing to unpack ../motd-news-config_14ubuntu6.1_all.deb...
Unpacking motd-news-config (14ubuntu6.1) over (14ubuntu6)...
Preparing to unpack ../base-files_14ubuntu6.1_amd64.deb...
Unpacking base-files (14ubuntu6.1) over (14ubuntu6)...
Setting up base-files (14ubuntu6.1)...
motd-news.service is a disabled or a static unit not running, not starting
(Reading database... 139606 files and directories currently installed.)
Preparing to unpack ../distro-info-data_0.68ubuntu0.1_all.deb...
Unpacking distro-info-data (0.68ubuntu0.1) over (0.68build1)...
Preparing to unpack ../snapd_2.75.2+ubuntu26.04.2_amd64.deb...
Unpacking snapd (2.75.2+ubuntu26.04.2) over (2.74.1+ubuntu26.04.4)...
Setting up motd-news-config (14ubuntu6.1)...
Setting up snapd (2.75.2+ubuntu26.04.2)...
Installing new version of config file /etc/apparmor.d/usr.lib.snapd.snap-d
snapd.failure.service is a disabled or a static unit not running, not star
snapd.gpio-chardev-setup.target is a disabled or a static unit not running
snapd.snap-repair.service is a disabled or a static unit not running, not
Setting up distro-info-data (0.68ubuntu0.1)...
Processing triggers for plymouth-theme-ubuntu-text (24.004.60+git20250831.
Processing triggers for dbus (1.16.2-2ubuntu4)...
Processing triggers for install-info (7.2-5ubuntu2)...
Processing triggers for plymouth (24.004.60+git20250831.4a3c171d-0ubuntu8)
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for dracut (110-11)...
update-initramfs: Generating /boot/initrd.img-7.0.0-15-generic
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

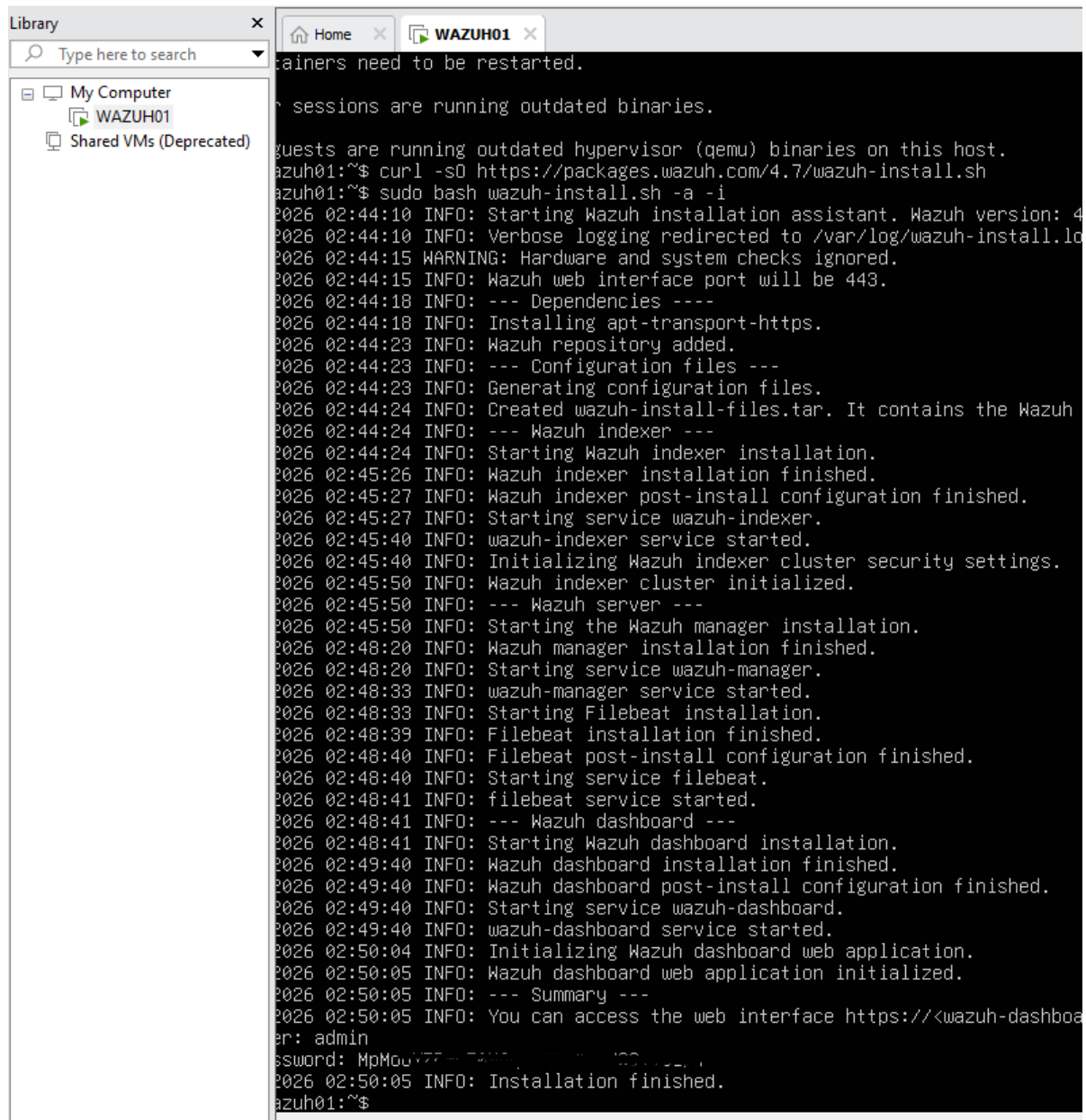
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
easy@wazuh01:~$ curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
easy@wazuh01:~$
```

Step 27 — Downloaded Wazuh Installer

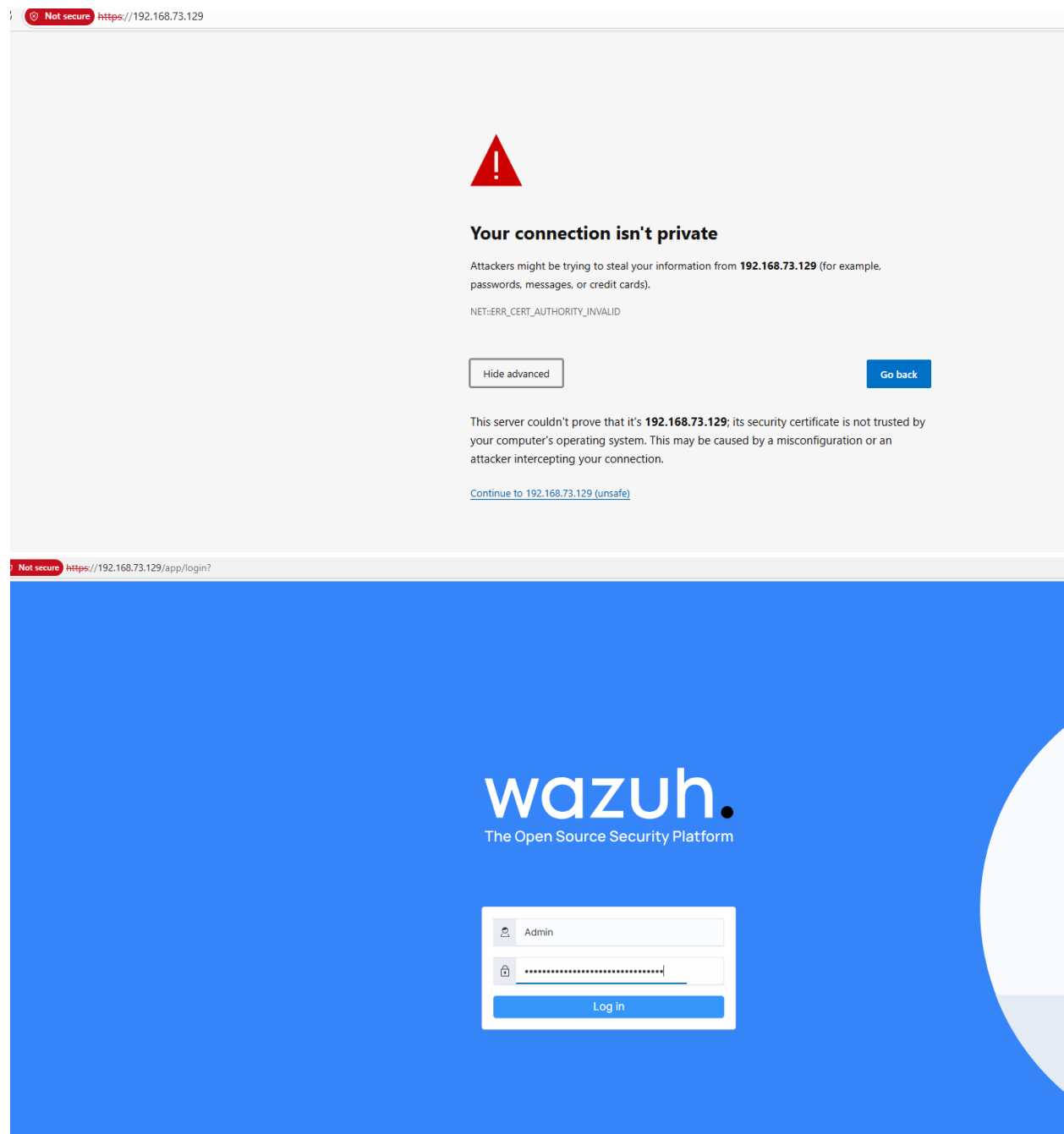
- Command: `curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh`
- Purpose: Downloaded Wazuh installation script



Step 28 — Wazuh Installation Complete

- Command: `sudo bash wazuh-install.sh -a -i`
- Wazuh Indexer: Installed 
- Wazuh Manager: Installed 
- Filebeat: Installed 
- Wazuh Dashboard: Installed 

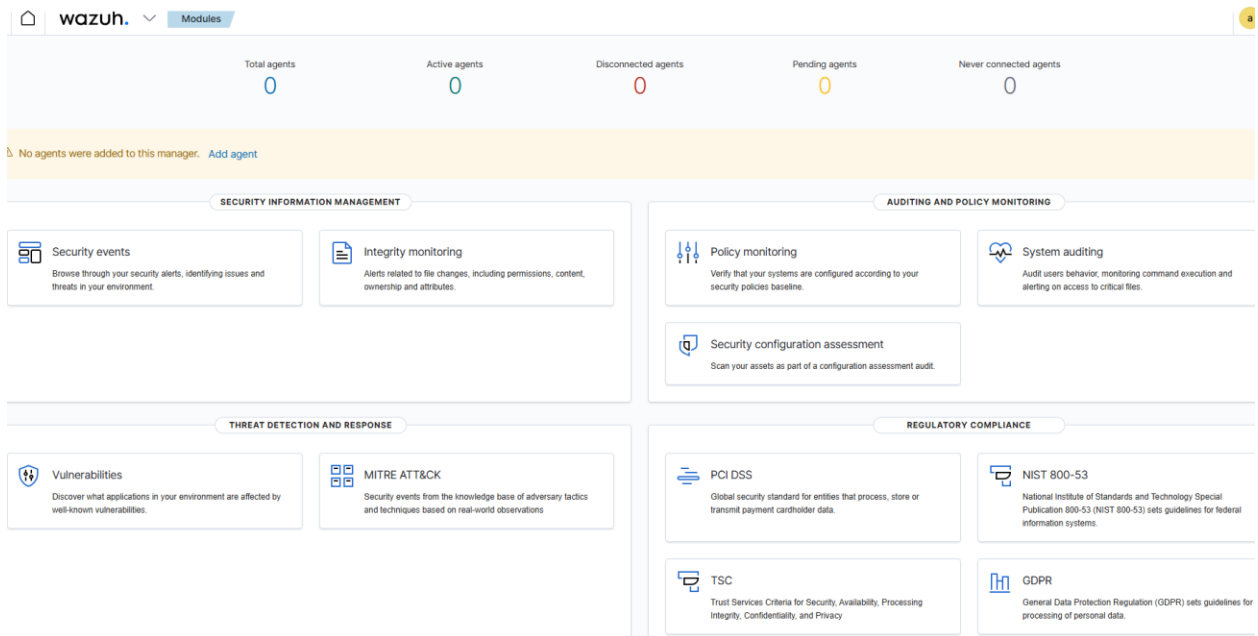
- Username: admin



Step 29 — Wazuh Dashboard Login Page

- URL: https://192.168.73.129

- Status: Dashboard accessible



Step 30 — Wazuh Dashboard Running

- Status: Fully operational
- Total Agents: 0 (no agents added yet)

Outcome:

WAZUH01 is fully built and operational.

The Wazuh SIEM dashboard is accessible at:

<https://192.168.73.129>

Next step: Add agents to start monitoring devices.

Lessons Learned:

1. Running VMs on external drives causes I/O errors and system instability. Always use internal storage.
2. Wazuh 4.7 requires the -i flag to install on

Ubuntu 26.04 as it is newer than supported versions.

3. Always run `sudo apt update` and `upgrade` before installing any software.